1 Chapter 3.1

Problem 2. *How many integers in* [100] *are not divisible by 4, 6, or 7?*

Answer:

$$\begin{aligned} N_{\geq}(\emptyset) &= 100 & N_{\geq}(d_4, d_6) &= \left\lfloor \frac{100}{2 \cdot 6} \right\rfloor = 8 \\ N_{\geq}(d_4) &= \left\lfloor \frac{100}{4} \right\rfloor = 25 & N_{\geq}(d_4, d_7) &= \left\lfloor \frac{100}{4 \cdot 7} \right\rfloor = 3 \\ N_{\geq}(d_6) &= \left\lfloor \frac{100}{6} \right\rfloor = 16 & N_{\geq}(d_6, d_7) &= \left\lfloor \frac{100}{6 \cdot 7} \right\rfloor = 2 \\ N_{\geq}(d_7) &= \left\lfloor \frac{100}{7} \right\rfloor = 14 & N_{\geq}(d_4, d_6, d_7) &= \left\lfloor \frac{100}{2 \cdot 6 \cdot 7} \right\rfloor = 1 \end{aligned}$$

The answer is $100 - (25 + 16 + 14) + (8 + 3 + 2) - 1 = 57$

Problem 6. *Answer the hat-check problem (i.e., the problem of derangement) for general n. This number is known as D_n. The question is "How many ways can the hats of n people be re-distributed so that each person receives exactly one hat, but no person receives there own hat"?*

Answer:
Before diving into the general n solution. Lets look at an example party problem, with 3 people, called Kaelin, Alyse and Donna. The three cases to avoid are the following:

$$\begin{aligned} h_k &:= \text{“Kaelin gets his own hat back”} \\ h_a &:= \text{“Alyse gets her own hat back”} \\ h_d &:= \text{“Donna gets her own hat back”} \end{aligned}$$

So the set $P = \{h_k, h_a, h_d\}$ and the possible subsets of the problem are:

$$J = \{\emptyset, \{h_k\}, \{h_a\}, \{h_d\}, \{h_k h_a\}, \{h_k h_d\}, \{h_a h_d\}, \{h_k h_a h_d\}\}$$

Instead of looping through each element in J and adding them individually, a better approach would be to group cases where the same amount of people receive there own hat back. For instance.

- $j = 0$, at least no one gets there hat back, $\emptyset$, 1 option
- $j = 1$, at least one person gets there hat back, $\{h_k\}, \{h_a\}, \{h_d\}$, 3 options
- $j = 2$, at least two persons gets there hat back, $\{h_k h_a\}, \{h_k h_d\}, \{h_a h_d\}$, 3 options
- $j = 3$, at least three persons gets there hat back, $\{h_k h_a h_d\}$, 1 option

Each of the number of options can be representented as "With 3 people, how many ways can j people get there hat back"? This is represented as $\binom{3}{j}$ or $\binom{n}{j}$ for a general amount of people, n.

Now we look at how to distribute the hats to people. If we didn't need to fix anyone getting there hat, the ways to distribute them would be 3!. Kaelin would have 3 options, Alyse would have 2 options and Donna would have 1 option. If instead we fixed one person, say Kaelin, then you only have 2 options to choose from, 2!. This is represented mathematically as $(3 - j)!$ or $(n - j)!$ for the general case.

Putting this together for the 3-person example we have:

$$D_3 = \sum_{j=0}^{3} \binom{3}{j} (-1)^j (3 - j)!$$

Where the $(-1)^j$ is the book keeping for ensuring odd values are subtracted and evens added. This prevents accidental double counting of values that appear in multiple cases. For the general case, where its n people, the equation is:

$$D_n = \sum_{j=0}^{n} \binom{n}{j} (-1)^j (n-j)!$$

Algebraically, this can be simplified starting with substiting the combination for its formula.

$$D_n = \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} (-1)^j (n-j)! = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}$$

Problem 7. *The number D_n is the hat check problem for general n.*

(a) Calculate $\lim_{n\to\infty} \frac{D_n}{n!}$

(b) Prove that for any n, D_n equals the closest integer to $n!/e$.

Answer:

(a) Looking at the limit of the hat check derangement over the total ways to distribute the hats.

$$\frac{D_n}{n!} = \frac{n!}{n!} \sum_{j=0}^{n} \frac{(-1)^j}{j!} = \sum_{j=0}^{n} \frac{(-1)^j}{j!}$$

The inverse of Euler's number can be expressed with the following infinite series.

$$\frac{1}{e} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!}$$

As $n \to \infty$ the summation approachs $1/e$. So the limit is the inverse of Euler's number.

(b) Explanation of the alternating series remainder term theorem from calculus here.

Before jumping right into using the alternating series remainder theorem, we need to first show the series converges. The criteria is:

$$\lim_{j\to\infty} \frac{1}{j!} = 0 \quad \text{and} \quad \frac{1}{(j+1)!} < \frac{1}{j!}$$

So it passes. We can now apply the alternating series remainder theorem. This says the error of evaluated term will be less than or equal to the value of the first term left off. When $j = 0$ or 1 the series value is 1, 0. This doesn't make sense to begin evaluating the series at these values. The first value that makes sense is $j = 2$, so two people.

$$\sum_{j=0}^{2} \frac{(-1)^j}{j!} = 1 - 1 + 1/2 = 1/2$$

The next term is $j = 3$ which gives $1/6$. So the low and high side of the $j = 2$ case is $1/2 \pm 1/6 = (2/6, 4/6)$ or $(1/3, 2/3)$. The value this finally converges to is $1/e = 1.1/3$ which falls inside the boundary.

Proof. For any value of n the number of derangements is:

$$D_n = n! \sum_{j=0}^{n} \frac{(-1)^j}{j!}$$

The number of derangements over total permutations $D_n/n!$ is a converging series, as it meets the alternating series theorem criteria.

$$\lim_{j\to\infty}\frac{1}{j!}=0 \quad \text{and} \quad \frac{1}{(j+1)!}<\frac{1}{j!}$$

This series converges to $1/e$ as n approaches infinity.

$$\frac{D_n}{n!}=\sum_{j=0}^{\infty}\frac{(-1)^j}{j!}=\frac{1}{e}$$

The alternating series remainder theorem states the max difference between the summation evaluted at n and the final converged value will be less than the single valve at $n+1$. For the probability of derangements this appears as

$$\sum_{j=0}^{n}\frac{(-1)^j}{j!}-\frac{1}{e}\leq\frac{1}{(n+1)!}$$

Multiplying through by $n!$ for number of derangements.

$$n!\sum_{j=0}^{n}\frac{(-1)^j}{j!}-\frac{n!}{e}\leq\frac{n!}{(n+1)n!}=\frac{1}{n+1}$$

As such the difference between D_n and $n!/e$ will always be less than an integer, allowing rounding the irrational number to provide an accurate count. □

Problem 8. *How many ways can you distribute 20 identical objects to 10 distinct recipients so that each recipient receives at most five objects?*

Answer:
The empty set at least (base case) is how many ways can you distribute 20 identical objects to 10 distinct recipients. This is

$$N_{\geq}(\emptyset)=\left(\!\!\binom{10}{20}\!\!\right)$$

Then you need to evaluate the following cases which are:

(a) P_1 - 1 person receives 6 or more objects

(b) P_2 - 2 people receives 6 or more objects

(c) P_3 - 3 people receives 6 or more objects

Taking the first case, P_1. Out of the 10 recipients, you can select 1 recipient to receive at least 6 objects. mathematically this is $\binom{10}{1}$, then you need to distribute the remaining 14 objects to the 10 recipients. Those objects can go to any person, all that matters is that at least 1 recipient has at least 6 objects, $\left(\!\binom{10}{14}\!\right)$.

$$N_{\geq}(P_1)=\binom{10}{1}\left(\!\!\binom{10}{14}\!\!\right)$$

Second case P_2, now two recipients get at least 6 objects. So you have 8 objects left over to distribute to the ten recipients.

$$N_{\geq}(P_2)=\binom{10}{2}\left(\!\!\binom{10}{8}\!\!\right)$$

Third case P_3, now three recipients get at least 6 objects. So you have 2 objects left over to distribute to the ten recipients.

$$N_{\geq}(P_3)=\binom{10}{3}\left(\!\!\binom{10}{2}\!\!\right)$$

The total answer is:

$$\left(\!\!\binom{10}{20}\!\!\right) - \binom{10}{1}\left(\!\!\binom{10}{14}\!\!\right) + \binom{10}{2}\left(\!\!\binom{10}{8}\!\!\right) - \binom{10}{3}\left(\!\!\binom{10}{2}\!\!\right)$$

Problem 9. *How many ways can you distribute k identical objects to n distinct recipients so that each recipient receives at most r objects?*

Answer: I just took problem 8 and make it a general case for problem 9.

$$\sum_{j=0}^{k/(r+1)} (-1)^j \binom{n}{j} \left(\!\!\binom{n}{k - j(r+1)}\!\!\right)$$

Problem 17. *A taxi drives from the intersection labeled A to the intersection labeled B in the grid of streets shown on page 94. The driver can only drive north (up) or east (right). Traffice reports indicate that there is heavy congestion at the intersections identified. How many routes from A to B can the driver take that . . .*

(a) Have no restrictions

(b) Avoid all the congested intersections?

(c) Pass through at most one congested intersection?

Answer:

(a) To get from A to B requires moving across 16 points. Each one of those points you can make two decisions, either move right 1 or move up 0. To reach your desired endpoint requires you to make exactly 10 right moves or exactly 6 up moves. So you have this many paths to choose from:

$$\binom{16}{10} = \binom{16}{6} = 8008$$

Imagine you have a bag of labels that look like, where if you select a label, it means you will move right on that move. So m_1 implies you move right on the first move. If you don't select the label, you need to move up.

$$\{m_1, m_2, m_3, \ldots m_{16}\}$$

Because you have defined the moves in terms of right moves, you need to pick 10 for a fully defined sequence. So the following would be a good example, which creates a zig zag pattern initially.

$$\{m_1, m_3, m_5, m_7, m_9, m_{11}, m_{13}, m_{14}, m_{15}, m_{16}\}$$

(b) Need to describe the four congested intersections. They will be described as the total number of moves, I_{jt} and number of right moves, I_{jr}. The number of ups required is $I_{ju} = I_{jt} - I_{jr}$

(i) $I_a :=$ Total: 5, rights: 2, ups: 3, Math $\binom{5}{2} = 10$

(ii) $I_b :=$ Total: 5, rights: 4, ups: 1, Math $\binom{5}{4} = 5$

(iii) $I_c :=$ Total: 9, rights: 7, ups: 2, Math $\binom{9}{7} = 36$

(iv) $I_d :=$ Total: 14, rights: 9, ups: 5, Math $\binom{14}{9} = 2002$

The set $P = \{I_a, I_b, I_c, I_d\}$ and the possible subsets is the power set of P, $\mathscr{P}(P)$. A secondary intersection, I_2, can be accessed from a primary intersection, I_1 only if the following is true.

$$I_{2r} > I_{1r} \text{ and } I_{2u} > I_{1u}$$

The secondary intersection requires more right moves and more up moves than the primary intersection. With that being said, it is impossible to access I_c or I_b from I_a. These combinations should be removed. So the number of possible movement sequences:

$$J = \{\emptyset, \{I_a\}, \{I_b\}, \{I_c\}, \{I_d\}, \{I_a I_d\}, \{I_b I_c\}, \{I_b I_d\}, \{I_c I_d\}, \{I_b I_c I_d\}\}$$

As seen, no subset with both $I_a I_b$ or $I_a I_c$ is occuring inside J. To mathematically account for landing on a secondary intersection after a primary intersection. You need to subtract the number of moves required to reach the primary intesection from the secondary intersection. If only landing on one intersection, you still need to account for finishing at B, so the total possible moves is:

$$N_{\geq}(I_a) = \binom{5}{2}\binom{16-5}{10-2} = 1650 \qquad N_{\geq}(I_b) = \binom{5}{4}\binom{16-5}{10-4} = 2310$$
$$N_{\geq}(I_c) = \binom{9}{7}\binom{16-9}{10-7} = 1260 \qquad N_{\geq}(I_d) = \binom{14}{9}\binom{16-14}{10-9} = 4004$$

Calculating the double intersection sequences:

$$N_{\geq}(I_a I_d) = \binom{5}{2}\binom{14-5}{9-2}\binom{16-14}{10-9} = 720 \qquad N_{\geq}(I_b I_c) = \binom{5}{4}\binom{9-5}{7-4}\binom{16-9}{10-7} = 700$$
$$N_{\geq}(I_b I_d) = \binom{5}{4}\binom{14-5}{9-4}\binom{16-14}{10-9} = 1260 \qquad N_{\geq}(I_b I_d) = \binom{9}{7}\binom{14-9}{9-7}\binom{16-14}{10-9} = 720$$

Calculating the lone triple intersection sequence:

$$N_{\geq}(I_b I_c I_d) = \binom{I_{bt}}{I_{br}}\binom{I_{ct}-I_{bt}}{I_{cr}-I_{br}}\binom{I_{dt}-I_{ct}}{I_{dr}-I_{cr}}\binom{A_t - I_{dt}}{A_r - I_{dr}} = \binom{5}{4}\binom{9-5}{7-4}\binom{14-9}{9-7}\binom{16-14}{10-9}$$

As such $N_{\geq}(I_b I_c I_d) = 5 \cdot 4 \cdot 10 \cdot 2 = 400$. The final answer is:

$$N_{=}(\emptyset) = 8008 - (1650 + 2310 + 1260 + 4004) + (720 + 700 + 1260 + 720) - 400 = 1784$$

(c) The clean answer for the problem is:

$$N_{\leq}(1) = N_{=}(\emptyset) + N_{=}(1)$$

We already have $N_{=}(\emptyset)$ from part b. We now need to compute $N_{=}(1)$ and add it to part b. $N_{=}(1)$ has four separate parts.

$$N_{=}(1) = N_{=}(I_a) + N_{=}(I_b) + N_{=}(I_c) + N_{=}(I_d)$$

Each part of $N_{=}(1)$ is calculated with the general inclusion-exclusion formula. Where:

$$N_{=}(S) = \sum_{J:S\subseteq J\subseteq P} (-1)^{|J|-|S|} N_{\geq}(J)$$

The addition of $|S|$ in the general inclusion-exclusion formula swaps the signs of $+$ or $-$:

$$\begin{array}{llllllll}
N_{=}(I_a) = N_{\geq}(I_a) & - N_{\geq}(I_a I_d) & & & & \\
N_{=}(I_b) = N_{\geq}(I_b) & & - N_{\geq}(I_b I_c) & - N_{\geq}(I_b I_d) & & + N_{\geq}(I_b I_c I_d) \\
N_{=}(I_c) = N_{\geq}(I_c) & & - N_{\geq}(I_b I_c) & & - N_{\geq}(I_c I_d) & + N_{\geq}(I_b I_c I_d) \\
N_{=}(I_d) = N_{\geq}(I_d) & - N_{\geq}(I_a I_d) & & - N_{\geq}(I_b I_d) & - N_{\geq}(I_c I_d) & + N_{\geq}(I_b I_c I_d)
\end{array}$$

If you generalize this, it becomes obvious the double terms appear twice and the triple terms appear three times. As such.

$$N_{=}(1) = (1650 + 2310 + 1260 + 4004) - 2 \cdot (720 + 700 + 1260 + 720) + 3 \cdot 400 = 3624$$

In conclusion, $N_{=}(1) = 1784 + 3624 = 5408$